CAR DATA EDA REPORT

Summary

This report analyzes a dataset of 8,128 used cars with 12 key features, including manufacturing year, selling price, kilometers driven, fuel type, transmission type, ownership history, mileage, engine size, and maximum power. After handling missing values using median and mean imputation and removing outliers with the IQR method, the cleaned dataset provided a reliable foundation for meaningful analysis.

The visualizations revealed that newer cars, those with higher engine capacity or maximum power, and cars with automatic transmission tend to have higher selling prices. Petrol and diesel cars dominate the market, with diesel vehicles usually priced slightly higher. Mileage showed a weaker effect on price compared to performance-related features. Overall, the analysis highlights engine specifications, car age, and transmission type as the strongest predictors of used car prices.

Introduction

This report analyzes a car dataset containing 8128 rows and 12 columns. The dataset includes important information about used cars, such as the year of manufacture, selling price, kilometers driven, fuel type, seller type, transmission type, number of previous owners, mileage, engine size, and maximum power.

These features help us understand how different car characteristics affect selling prices. By exploring and summarizing this data, we can identify key trends, uncover pricing patterns, and gain insights that can support decision-making for buyers, sellers, and car dealerships.

In addition to understanding the structure of the dataset, it is important to examine the overall distribution of its key numerical features. Summary statistics such as the mean, median, minimum, maximum, and standard deviation provide a quick overview of the central values and variability within the data. These statistics help identify typical ranges, potential anomalies, and the general spread of each variable before moving into deeper analysis.

Data Cleaning

Before analyzing the data, several cleaning steps were performed to ensure accuracy and consistency. Missing values in key numeric columns were handled by replacing them with appropriate statistical measures: the *seats* and *engine* columns were filled using their median values, while *max_power* was first converted to numeric and then filled using its mean. Missing *mileage* values were also replaced with the median to maintain consistency across the dataset.

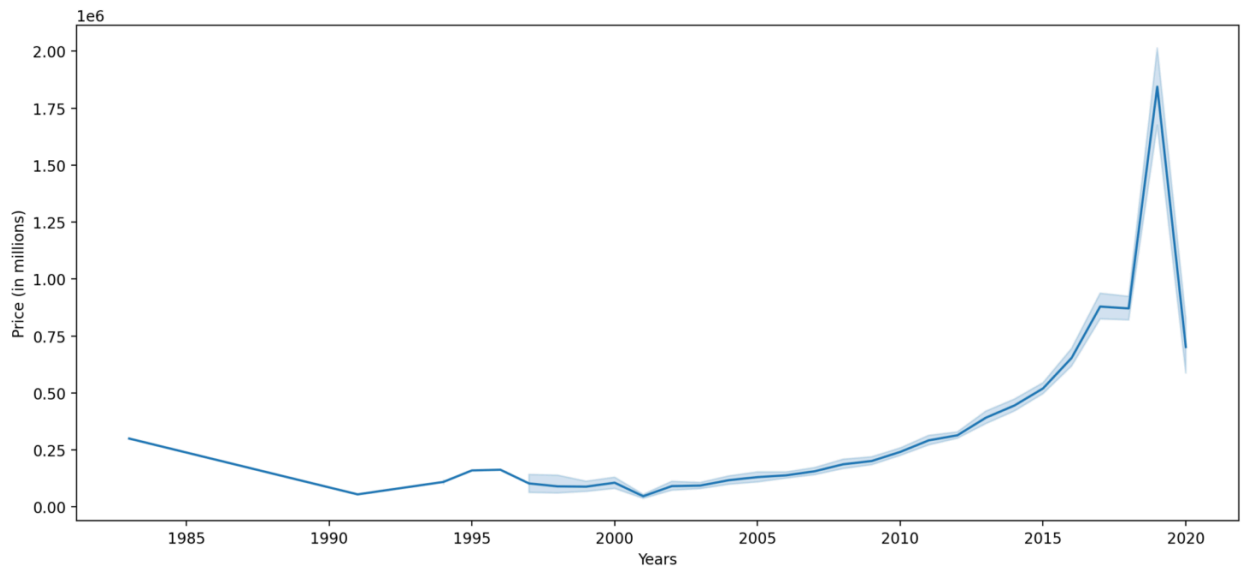
Additionally, numerical columns were checked for outliers using the Interquartile Range (IQR) method. For each numeric feature, values falling outside the range of **Q1 − 1.5×IQR and Q3 +**

1.5×IQR were removed. This helped reduce the effect of extreme values that could distort the analysis.

After these steps, the cleaned dataset was used for further visualization and exploration.

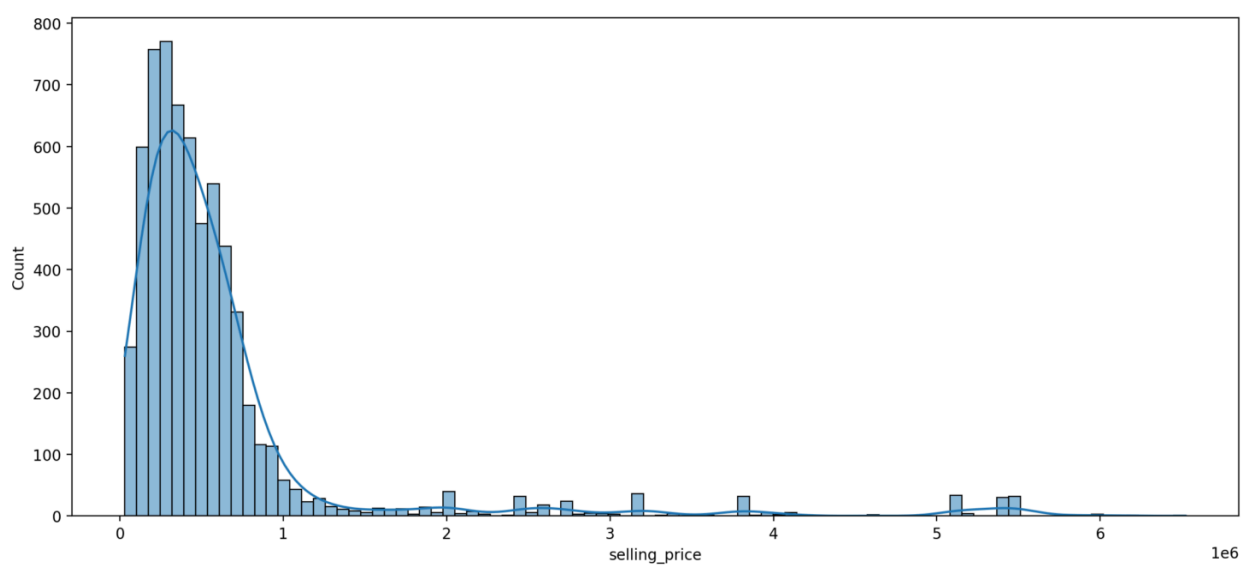
Data Visualization and Findings

Selling Price Over the Years



A line plot revealed that newer cars generally have higher selling prices, which is expected due to less wear and better features in recent models.

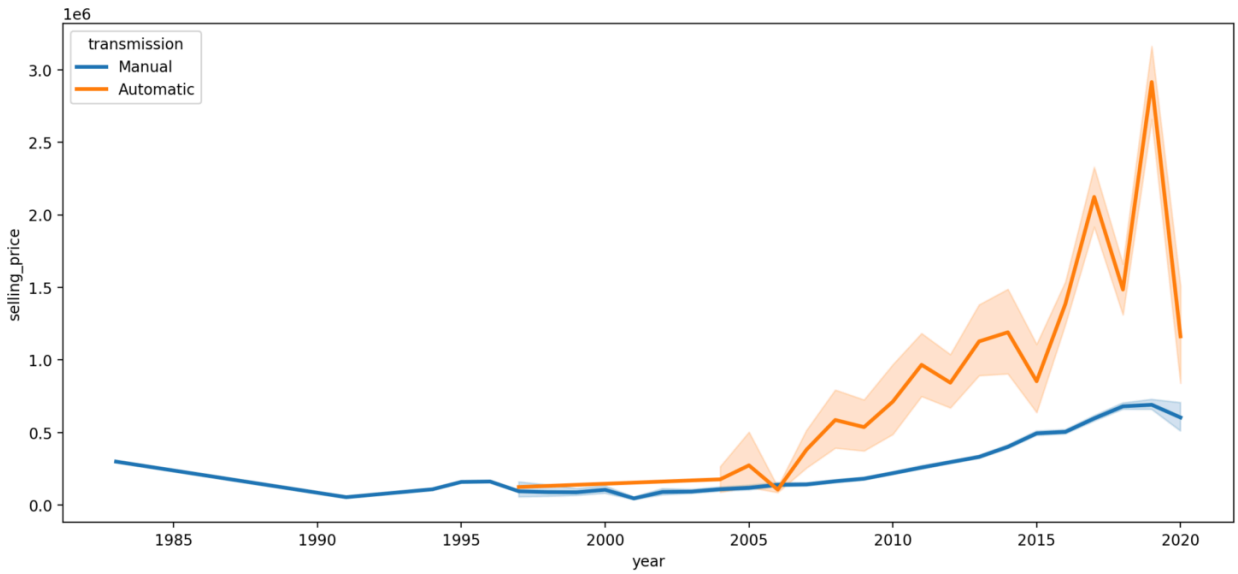
Distribution of Selling Price



The price distribution is right-skewed, meaning:

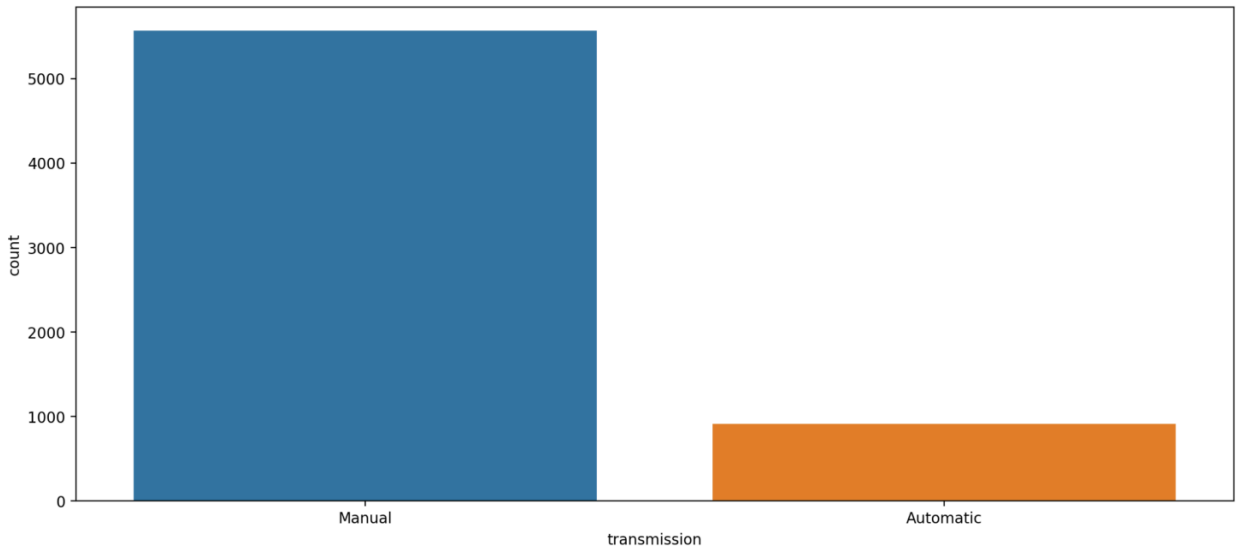
- Most cars are moderately priced
- A few luxury cars increase the upper price range significantly

Selling Price by Transmission Type



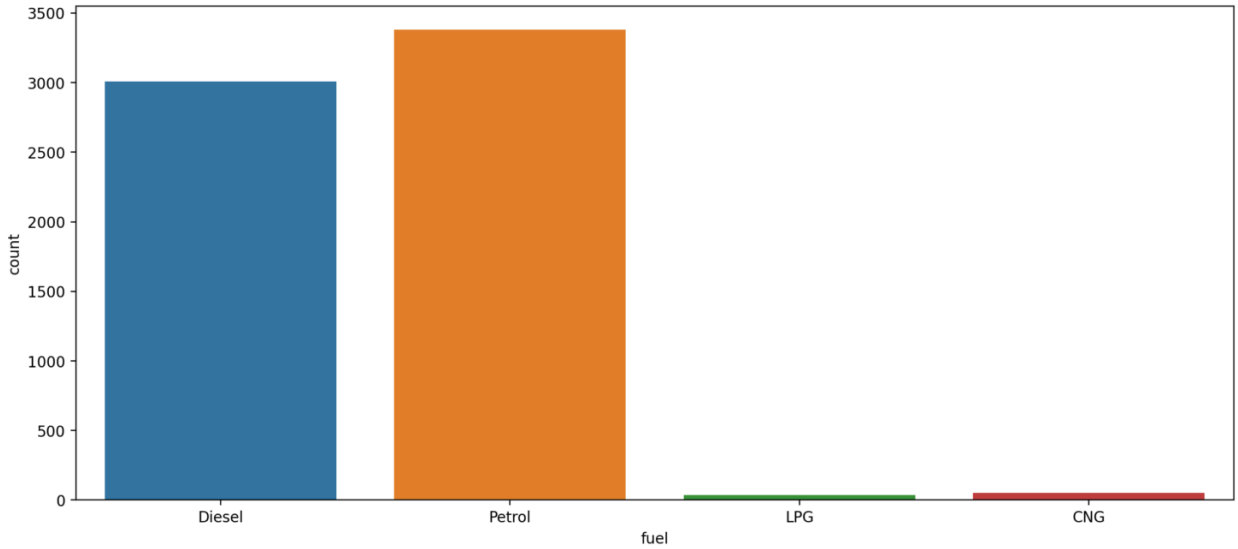
Vehicles with automatic transmission consistently showed higher selling prices than manual ones. This suggests higher perceived value or demand for automatic cars.

Transmission Type Distribution



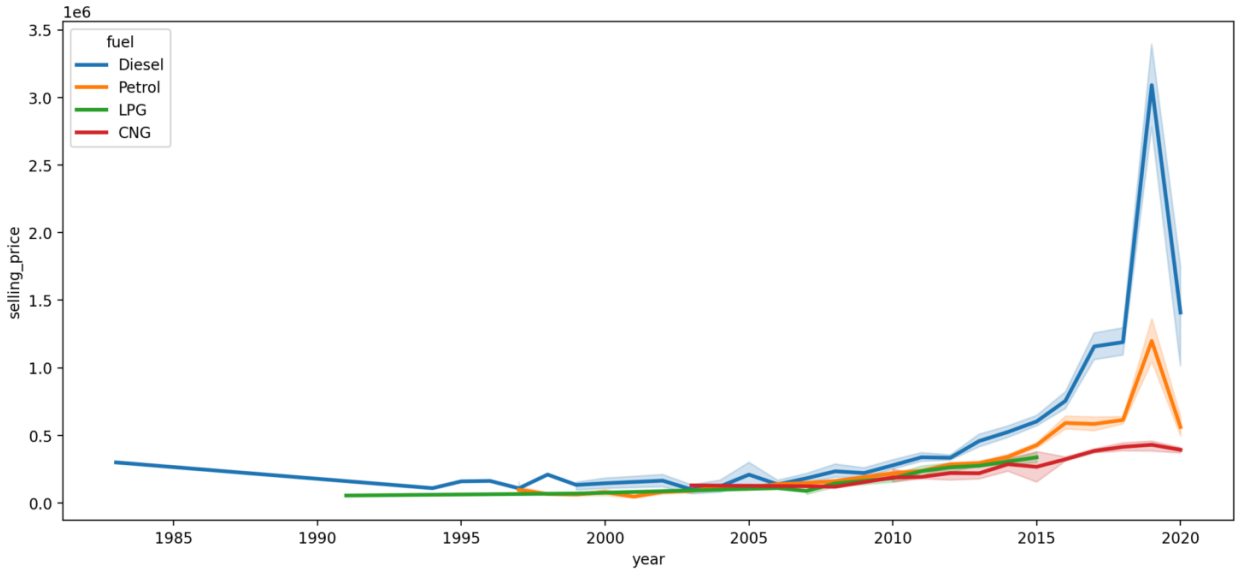
Manual transmission accounts for most listings, but automatic transmission is steadily increasing in availability in recent years.

Distribution of Fuel Types



Fuel type analysis showed that petrol and diesel cars are the most common, with CNG and LPG making up a very small proportion.

Selling Price by Fuel Type

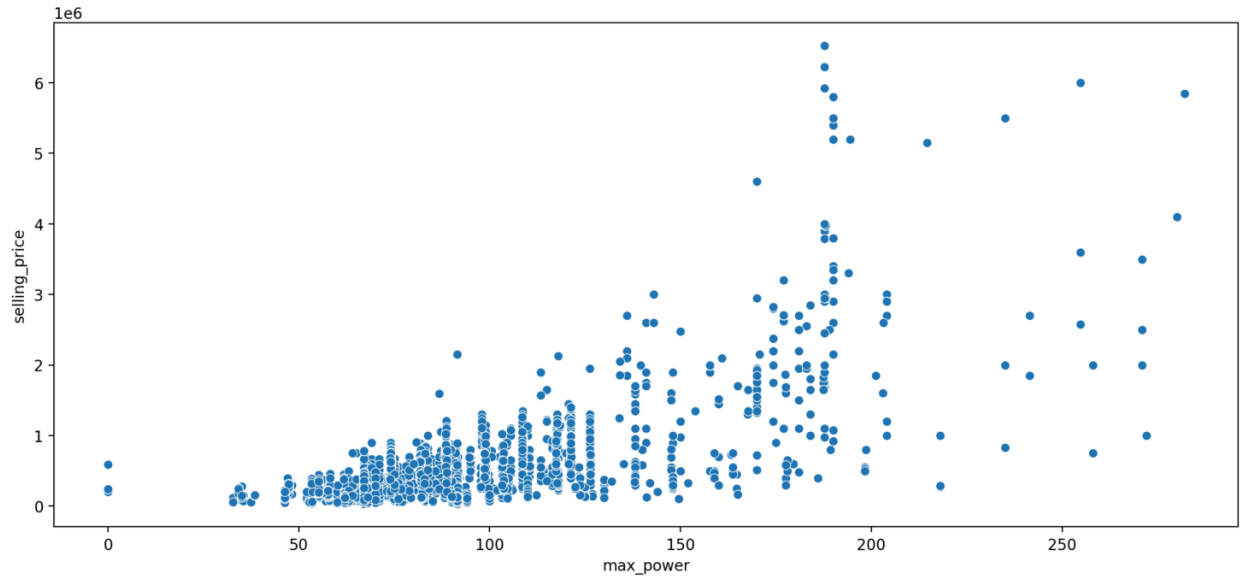


Diesel cars tend to have highest selling prices throughout years even though more petrol cars were been sold, while petrol cars gained value in more recent years.

LPG cars seem to stopped selling around 2015, with CNG cars which started selling in the early 2000s still not been able to catch up with either diesel or petrol cars.

This shows the cost effectiveness of the diesel cars throughout the years.

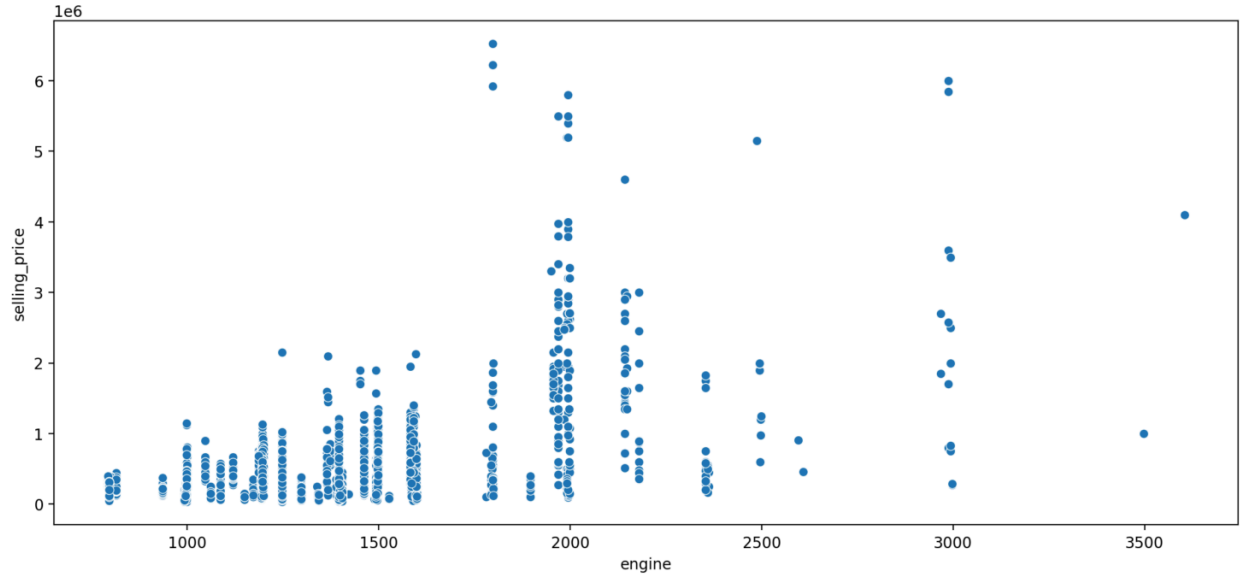
Maximum Power vs Selling Price



A scatter plot confirmed a positive relationship between maximum power and selling price:

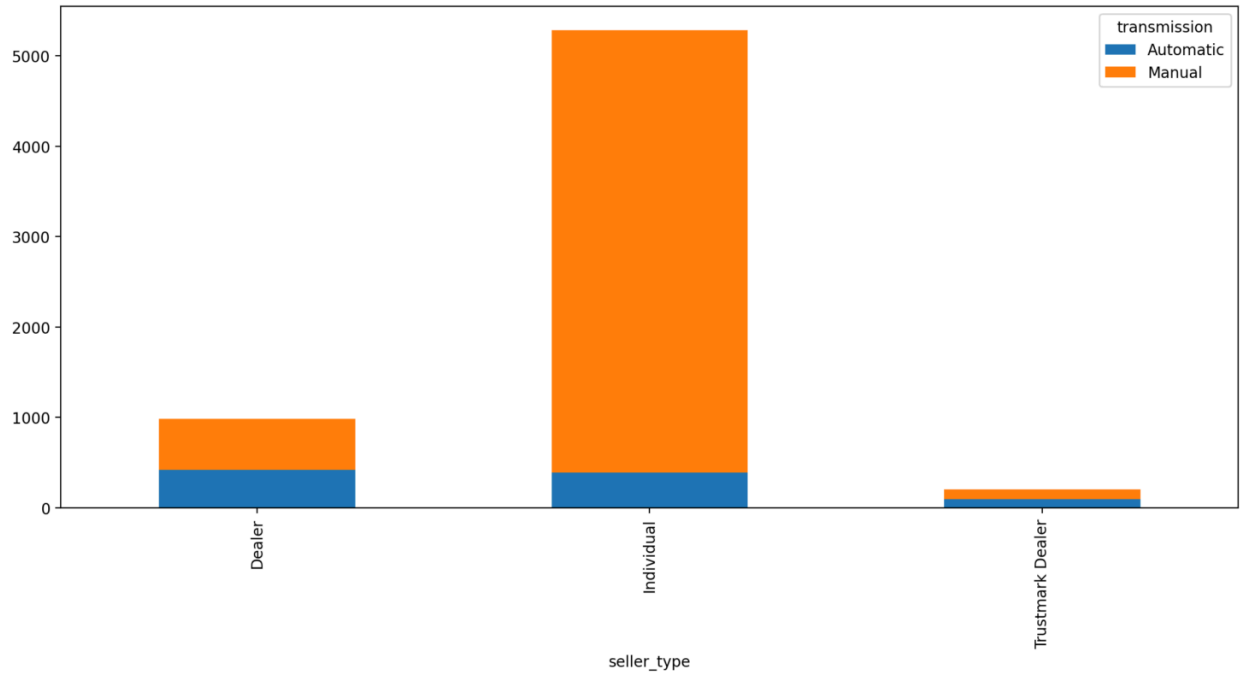
Cars with higher horsepower tend to be more expensive.

Engine Size vs Selling Price



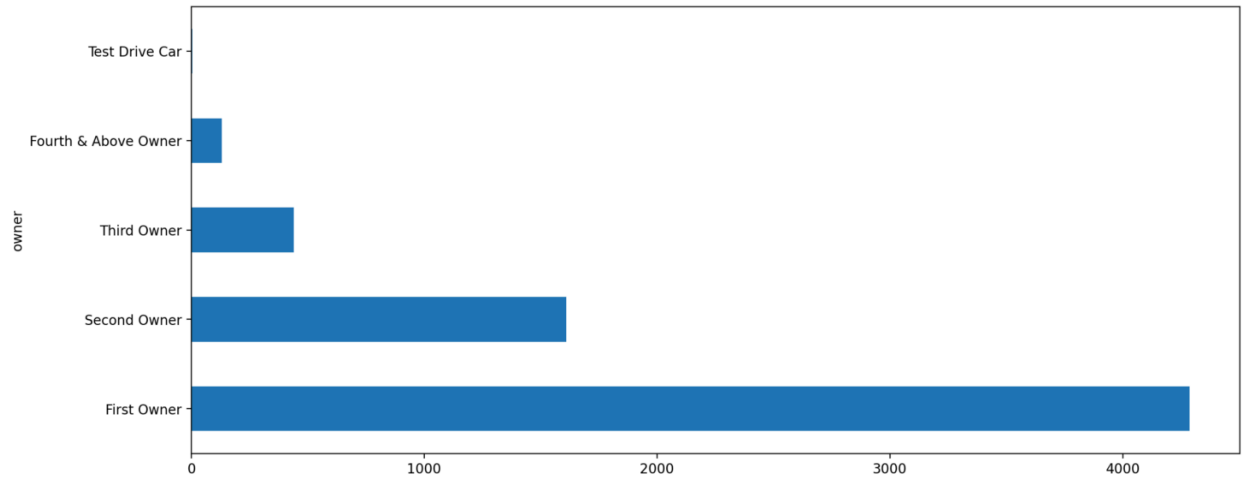
Larger engine sizes generally lead to higher car prices. The relationship is clear but contains outliers, showing some cars with modest engines being sold at higher prices due to other factors (brand, features, condition).

Seller Type vs Transmission



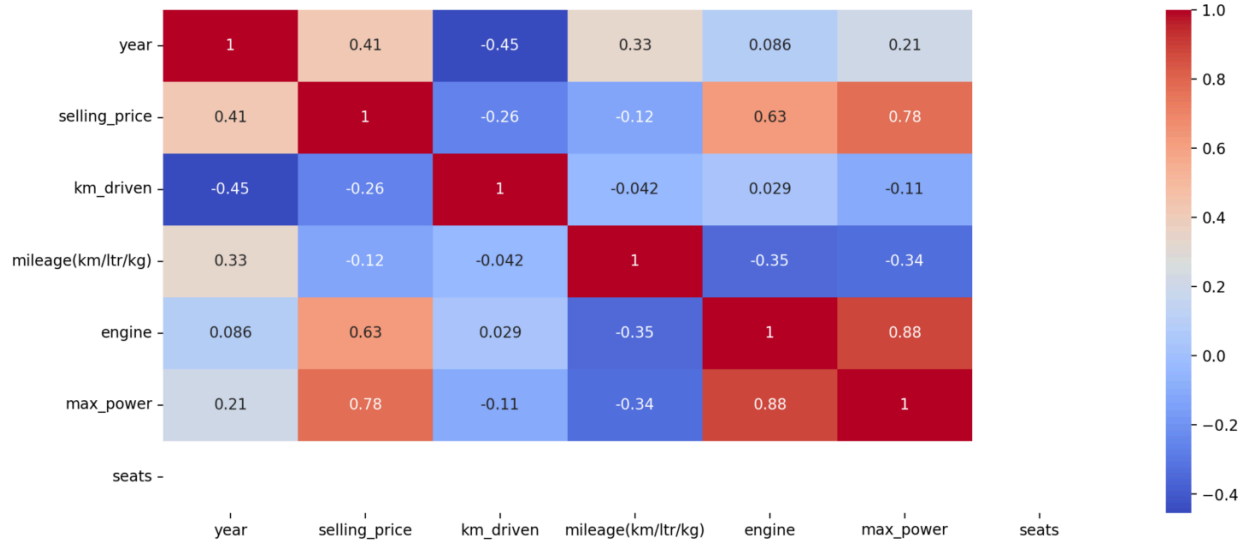
A stacked bar chart showed that individual sellers mostly list manual vehicles, while dealers list a higher proportion of automatic cars.

Car Ownership Distribution



The majority of cars are first-owner and second-owner vehicles. Third-owner and fourth-owner cars show significantly fewer listings, whiles test-drive cars show almost nothing.

Correlation Heatmap

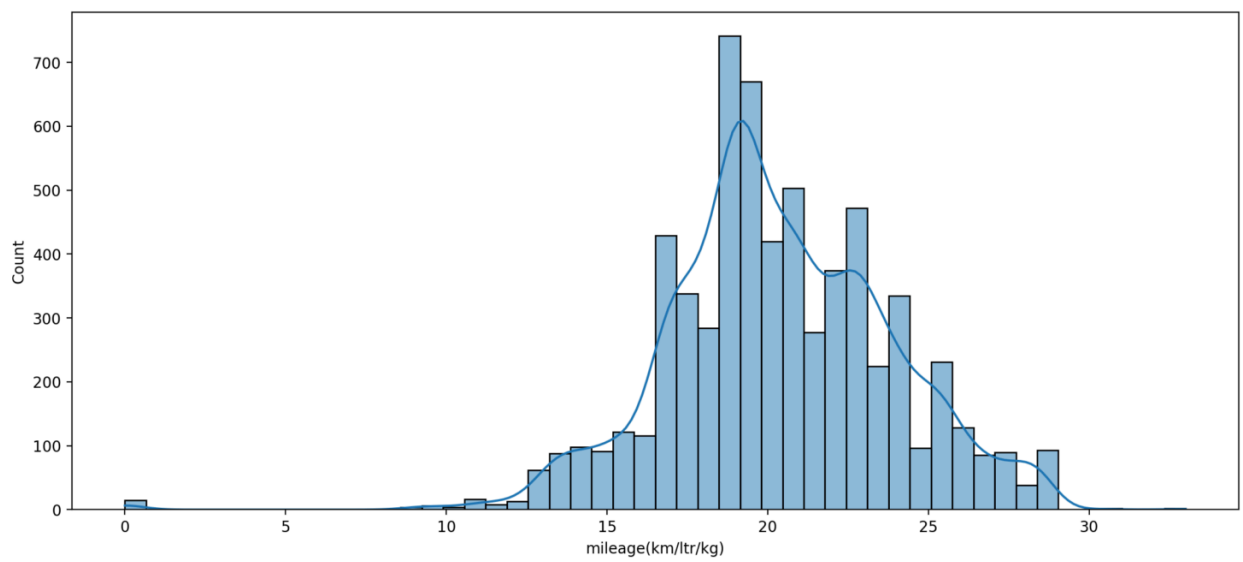


The correlation map highlighted:

- High correlation between *max_power* and *engine size*
- Positive correlation between *max_power* and *selling_price*
- Weak correlation between *km driven* and *selling price*

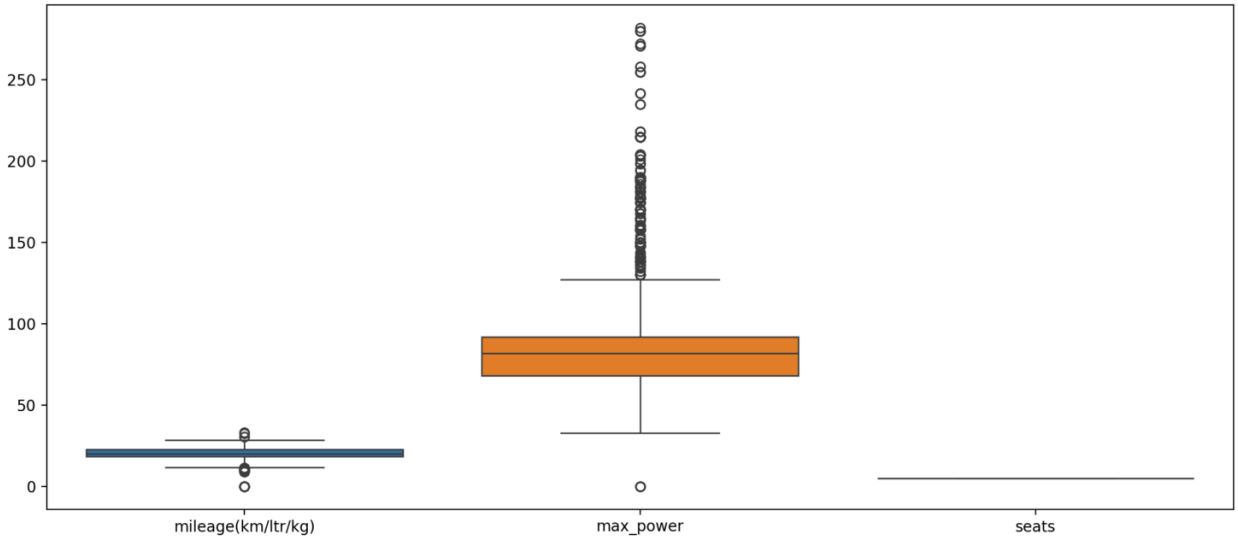
This suggests performance-related features are more important than mileage alone.

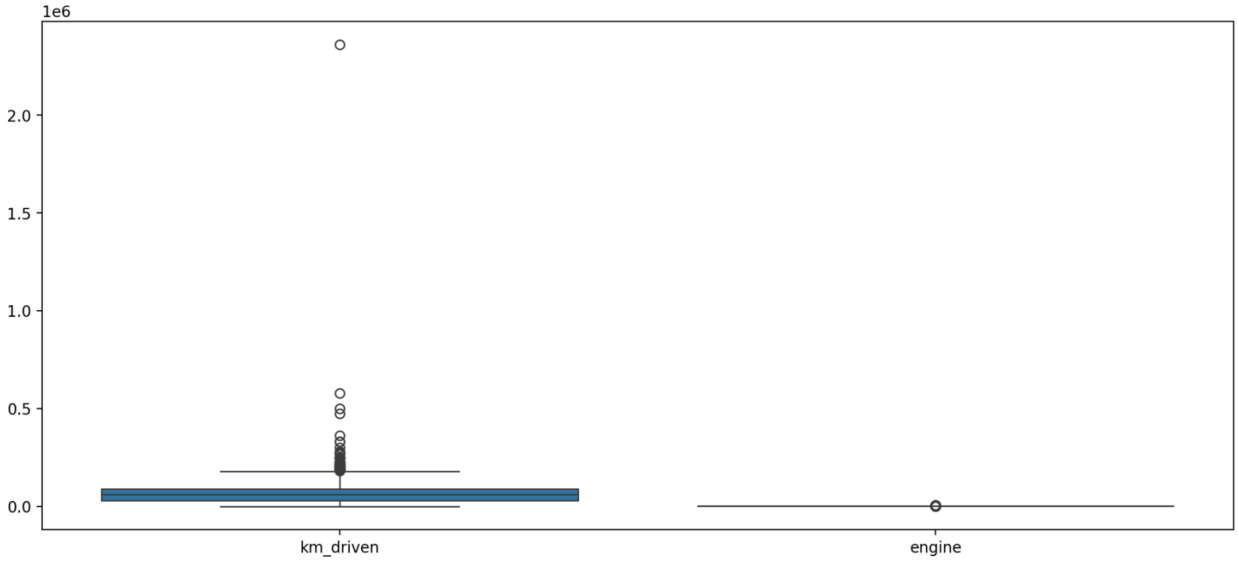
Distribution of Mileage

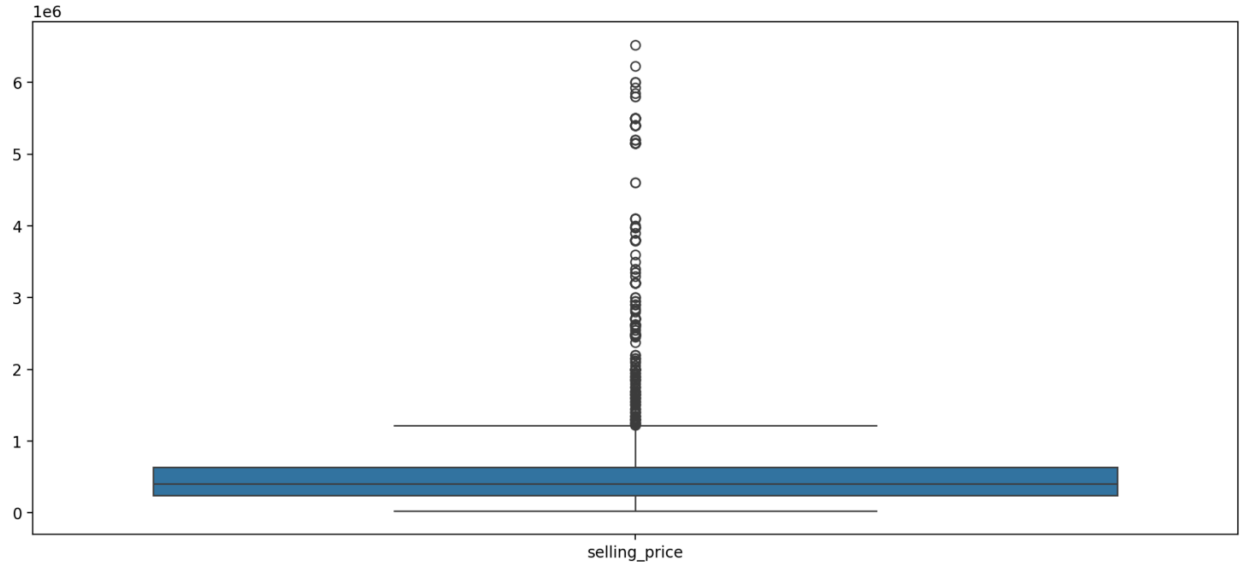


Mileage distribution shows that most cars fall within a moderate mileage range, with few extreme outliers (very high mileage cars).

Boxplots for Numeric Variables


250
200
150
100
50
0
mileage(km/ltr/kg)
max_power
seats


1e6
2.0
1.5
1.0
0.5
0.0
km_driven
engine



Boxplots revealed several outliers across features like kilometer driven and maximum power, indicating variability in car conditions and models.

Key Insights

- **Manual vs Automatic: Transmission Plays a Significant Role**
  - Even though more manual cars are listed or been sold than automatic cars, prices of automatic cars soar higher than manual cars.

- **Fuel Type: Diesel and Petrol Cars Dominate**
  - Fuel type affects price patterns, but the market is still dominated by petrol and diesel with diesel cars selling a little higher than petrol cars.

- **Mileage Has Less Impact Than Expected**
  - Mileage showed a weaker correlation with price than features like power or engine capacity.

- **Engine and Power Strongly Influence Price**
  - Cars with higher *engine* capacity and *max_power* consistently command higher prices.

- **Newer Cars Cost Significantly More**
  - Year of manufacture was one of the strongest predictors of price.

Conclusions

This analysis of the car dataset provides a solid understanding of the factors influencing used car prices. The results clearly show that car power, engine size, manufacturing year, and transmission type are major pricing determinants. These insights can help car buyers make better purchasing decisions and help sellers optimize pricing strategies based on key features.